

Assignment Briefing

Shiyu Zhao

Department of Artificial Intelligence
Westlake University

Assignment briefing

Problem setup:

s1	s2	s3	s4	s5
s6	s7			s8
s9		s10	s11	s12

- 1 target area, 3 forbidden areas
- 12 states, each of which has 5 actions
- The reward setting is $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, and $r_{\text{target}} = 1$.
- The discount rate is $\gamma = 0.9$.

Assignment briefing

Pay attention to the following to avoid unnecessary mistakes:

- You **should** use the grid-world task specified in the figure. You **should not** use other tasks. (otherwise, **deduct** up to 20% grade)
- The forbidden areas are **not accessible** and do **not** correspond to states. If the agent attempts to enter, it will get bounced back to its previous state. (otherwise, **deduct** up to 20% grade)
- You **should** use the reward and discount rate specified before. You **should not** use other values. (otherwise, **deduct** up to 20% grade)
- Each state has 5 actions. You **should** use algorithms to learn which actions are good or bad. You **should not** manually eliminate some actions even though you think they are not good. (otherwise, **deduct** up to 20% grade)
- You **should** fully utilize the code I provide for, for example, plotting arrows or trajectories. It will save you time. You **should not** do the plotting from scratch by yourself. (otherwise, **deduct** up to 20% grade)

Five assignments:

- A report for each assignment.
- Each report is worth 20% of the final grade.
- Each assignment is the foundation of the next. So make sure to complete each assignment carefully.
- Each assignment takes about ? hours.

- Contents: The report must *elaborate what you have done*. It must be able to *demonstrate that you understand the algorithms well*. The report may consist of texts, equations, figures, code, etc.
- Format: You can use Word, Latex, Markdown, or other tools to write the report.
- Language: The report should be in English. (otherwise, **deduct** up to 40% grade)
- Length: There is not page limitation. However, an overlong report should be avoided.

Source code:

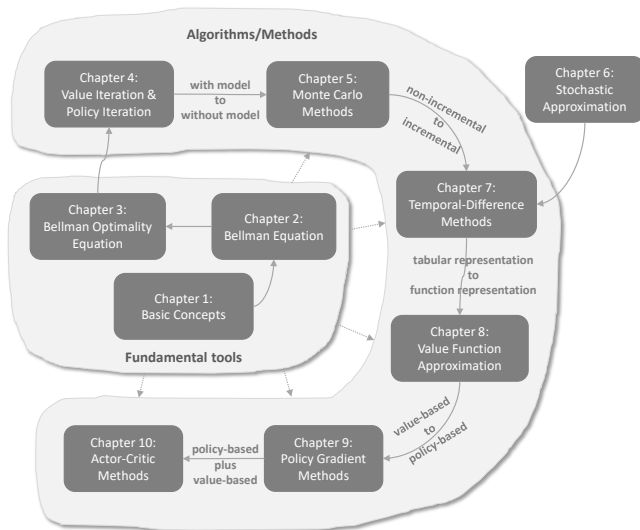
- The submitted code should be straightforward to run. A brief description of the usage of your code should be given in a text file. The code should be clearly structured and annotated. (otherwise, **deduct** up to 40% grade)
- You may choose what ever programming language you like. However, you should NOT use existing toolboxes or implementations of the above algorithms. Instead, you must develop your own algorithms from scratch. Preliminary operations such as matrix-vector and probability operations are allowed.

- Please submit via the online Canvas system. Do NOT submit via email to me.
- Please submit two files: the first is your report in PDF format, and the second is a zip file of your source code. The name of your report should be "*RL_student number_report*". The name of the zip file of your source code should be "*RL_student number_source code*".
- Please do NOT include your name anywhere in your submission. But make sure that you include your student number correctly!

- Time to submit: 23:59, the first or the second Monday right after the lecture
- Late submission: penalty of 20% mark per day

- Every student must work **independently** to complete the entire project. Discussion on certain technical problems among the students is allowed. However, note that this is not a group project. You should **NOT** work as a group to do the project.
- Plagiarism is prohibited. Once it is detected in the report or source code, it will be investigated carefully and the corresponding students will be punished accordingly.
- You are encourage to program by yourself. If there is obvious evidence showing that you use large language models to write the report or do the programming, we can **deduct** up to 100% grade of that assignment.

Assignment briefing



Assignment 1

▷ Purpose of this assignment:

- Build up the simulation environment.
- Understand the basic concepts in RL.

▷ The code about plotting is available! See the GitHub homepage.

This assignment is important. The code developed in this assignment will be frequently used in the following assignments.

Assignment 1

- ▷ In your report, you should clearly include the following contents.
- Describe the software that you use. (2.5% grade)
 - Describe your the reward setting and discount rate. (2.5% grade)
 - Design any two policies that you like. One should be deterministic and the other should be stochastic. Do the following for each policy.
 - Describe the policy using an array or table. (10%*2 grade)
 - Plot the policy: show arrows at each state to indicate the policy. (10%*2 grade)
 - Specify an arbitrary starting state that you like. Show the trajectory from the starting state following the policy. The length of the trajectory is 50 steps. (10%*2 grade)
 - Calculate and show the discounted return of this trajectory. (10%*2 grade)
- The above plot and calculation should be done automatically by your code. Some key parts of your code should be given and explained concisely. (10% grade)
- Discuss what you have observed about the two policies. (5% grade)

▷ Purpose of this assignment:

- Understand how to build up and solve the Bellman equation.

The code developed in this assignment will be used in the next assignment.

Assignment 2

- ▷ In your report, you should clearly include the following contents.
 - Describe the problem setup such as the task, environment, reward setting, and discount rate. (5% grade)
 - Select two policies that you like. One should be deterministic and the other should be stochastic. Do the following for each policy.
 - Plot the policy in the grid using arrows. (5%*2 grade)
 - Show the values of the vector r_π and the matrix P_π . (5%*2 grade)
 - Describe your understanding of the closed-form algorithm. Calculate the state values using the closed-form algorithm. Plot the values on the grid. (15%*2 grade)
 - Describe your understanding of the iterative algorithm. Calculate the state values using the iterative algorithm. Plot the values on the grid. (15%*2 grade)

The above plot and calculation should be done automatically by your code. Some key parts of your code should be given and explained concisely. (10% grade)

- Discuss what you have observed about the state values of the two policies and the two algorithms. (5% grade)

▷ Purpose of this assignment:

- Understand how to find optimal policies using the **value iteration** algorithm.

This is the first time that you can find optimal policies. You will use the code in the next assignment.

Assignment 3

- ▷ Your report should include the following contents.
- Describe the problem setup such as the task, environment, reward setting, and discount rate. (5% grade)
- Use the value iteration algorithm to do the following tasks:
 - Briefly describe your understanding of this algorithm. (5% grade)
 - Post the key parts of your code and make clear explanation about how you implement the algorithm. (35% grade)
 - Plot the obtained optimal policy and optimal state values in the grid. (20% grade)
 - Plot the evolvement of the values. That is to use the grid to visualize show how the values change from the initial one to the final optimal one. Please plot the first five and the last five figures if there are more than 10 iterations in your algorithm. Each figure should not be too big so that each page can show multiple figures. (20% grade)
- Discuss what you have observed in your experiments. (15% grade)

▷ Purpose of this assignment:

- Understand how to use off-policy Q-learning to find optimal policies.

The code developed in this assignment will be used in the next assignment.

Assignment 4

- ▷ In your report, you should clearly include the following contents.
 - Describe the problem setup such as the task, environment, reward setting, and discount rate. (5% grade)
 - Use the code that you developed in the last assignment to find and plot the optimal policy and optimal state values. They are used as references for comparison in this assignment. (10% grade)
 - Describe and plot the behavior policy that you use to collect experience samples. (10% grade)
 - Describe and plot the episode (or one of the multiple episodes) generated by the behavior policy. (10% grade)
 - Describe and plot the final policy you obtain. (10% grade)
 - Show the evolution process of the policies by plotting the state value error curve. (20% grade)
 - Post the key parts of your code and make clear explanation about how you implement the algorithm. (30% grade)
 - Summarize what you have observed in your experiments. (5% grade)

▷ Purpose of this assignment:

- Understand how to use off-policy Deep Q-learning to find optimal policies.

This is the final assignment.

Assignment 5

- ▷ In your report, you should clearly include the following contents.
- Describe the problem setup such as the task, environment, reward setting, and discount rate. You are encouraged to use the same setup as I used in my book. (5% grade)
 - Solve and plot the optimal policy and optimal state values based on the value iteration algorithm that you have developed in a previous assignment. (10% grade)
 - Describe and plot the behavior policy that you use to collect experience samples. Describe and show the episode (or one of the multiple episodes) generated by the behavior policy. (10% grade)
 - Describe the neural network you use. (10% grade)
 - Post the key parts of your code and make clear explanation about how you implement the algorithm. (40% grade)
 - Describe and show the final policy you obtain. (10% grade)
 - Show the loss function curve and the state value error curve. (10% grade)
 - Summarize what you have observed in your experiments. (5% grade)